

COURSE TITLE : ADVANCED MANUFACTURING TECHNOLOGY
COURSE CODE : 6302
COURSE CATEGORY : A
PERIODS / WEEK : 5
PERIODS / SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Plastic Moulding processes, Compression Moulding Unconventional Machining Processes	21
2	Electric Discharge Machines &ECM	19
3	Other non-conventional Machining	18
4	Computer Integrated Manufacturing, Rapid Prototyping	17
TOTAL		75

COURSE OUTCOME:

1. Understand the Plastic Moulding Process
2. Understand the Compression Moulding
3. Understand Unconventional Machining Processes
4. Analyze Electric Discharge Machines &ECM
5. Understand Different Non-Conventional Processes
6. Understand the Principles of Computer Integrated Manufacturing
7. Analyze Rapid Prototyping

MODULE I

1.1.0 Understand the Plastic Moulding Process

- 1.1.1 Describe Injection moulding
- 1.1.2 Describe Screw press & plunger type
- 1.1.3 Describe Operation & specification of moulding machines with Sketches

1.2.0 Understand the Compression Moulding

- 1.2.1 Describe Transfer moulding
- 1.2.2 Describe Extrusion moulding
- 1.2.3 Describe Blow moulding
- 1.2.4 Explain Laminating
- 1.2.5 Explain Calendaring
- 1.2.6 Explain vacuum forming

1.3.0 Understand Unconventional Machining Processes

- 1.3.1 Introduction
- 1.3.2 Describe the Types of energy utilization – Mechanical, Chemical, Electrical, Thermal and Thermo Electric Processes
- 1.3.3 Distinguish between Conventional & Unconventional machining

MODULE II

2.1.0 Analyze Electric Discharge Machines &ECM

- 2.1.1 Explain Concept of EDM, application and principles, advantages and disadvantages
- 2.1.2 Describe Machine type – contact initiated, spark initiated and die electric
- 2.1.3 Define Surface finish and accuracy
- 2.1.4 Describe EDM electrodes – metallic and non-metallic materials requirement of a good electrode material
- 2.1.5 Describe Dielectric fluid, need and requirement
- 2.1.6 Process description, material removal rate, polarity, pulse time, temperature, voltage and current during sparking, their effect on surface & MRR
- 2.1.7 Describe Spark gap and its importance, determination of spark gap, factors on which it depends.
- 2.1.8 Illustrate EDM machine construction, table, work head, motors, spindle, dielectric tank
- 2.1.9 Explain Flushing purpose and method of flushing
- 2.1.10 Understand Safety precautions
- 2.1.11 Explain Electro Chemical Machining (ECM) – principle – surface finish – application – advantages and limitations.
- 2.1.12 Describe Wire cut Electric Discharge Machining (WCEDM) – Introduction – principle – machining procedure – applications – advantages and limitations
- 2.1.13 Describe the Electro Chemical wire cutting.

MODULE III

3.1.0 Understand different Non-conventional processes

- 3.1.1 Describe Abrasive jet machining – introductions, equipment, accuracy and surface finish, applications, advantages and limitations
- 3.1.2 Explain Laser Beam Machining (LBM) – Introduction, equipment and machining procedure – accuracy – application advantages and limitations.
- 3.1.3 ULTRASONIC machining (USM) –sketch, method, advantages and disadvantages.
- 3.1.4 Explain Water Jet Machining (WJM) – Introduction – principle of operation – equipment – advantages – limitations & applications.
- 3.1.5 Explain Plasma Arc Machining – Introduction – Principle – accuracy and surface finish – advantages – limitations & applications.
- 3.1.6 Describe Electron Beam Machining – Introduction – Energy source – Principle – advantages – limitations – applications.
- 3.1.7 Describe Explosive fabrication – open type – closed type – multiple types

MODULE IV

4.1.0 Understand the principles of computer integrated manufacturing

- 4.1.1 Introduction
- 4.1.2 Illustrate CNC/DNC with line diagram
- 4.1.3 Explain Computer Integrated Manufacturing (CIM) -main activities under CIM

- 4.1.4 Explain Computer Aided Design (CAD) –components of a CAD system
- 4.1.5 Explain Computer Aided Manufacturing (CAM)- introduction – advantages of CAD and CAM application area of CAD and CAM
- 4.1.7 Describe In Brief Flexible manufacture system (FMS) –flexibility and automation- basic computer control system – human labor
- 4.1.6 Describe Types of flexible manufacturing system – flexible manufacturing unit (FMU) – Flexible Manufacturing cell (FMC) – flexible manufacturing transfer line (FMC) or (FTL)
- 4.2.0 Analyze Rapid Prototyping**
 - 4.2.1 Introduction to RP
 - 4.2.2 Describe Basic Principles of RP
 - 4.2.3 Explain RP Technologies (Liquid based, Solid based)
 - 4.2.4 Describe Stereo lithography
 - 4.2.5 Analyze Droplet Deposition manufacture
 - 4.2.6 Describe LOM
 - 4.2.7 Describe FDM
 - 4.2.8 Explain Practical applications of RP
 - 4.2.9 Enlist Limitations of RP

CONTENT DETAILS

MODULE I

Injection moulding-Screw press & plunger type-Operation & specification of moulding machines with Sketches
Transfer moulding-Extrusion moulding-Blow moulding-Laminating-Calendering-vacuum forming-Types of energy utilization – Mechanical, Chemical, Electrical, Thermal and Thermo Electric Processes-Difference between Conventional & Unconventional machining.

MODULE II

Electric Discharge Machines &ECM-Concept of EDM, application and principles, advantages and disadvantages-Machine type – contact initiated, spark initiated and die electric-Surface finish and accuracy-EDM electrodes – metallic and non-metallic materials requirement of a good electrode material. Dielectric fluid, need and requirement-Process description, material removal rate, polarity, pulse time, temperature, voltage and current during sparking, their effect on surface & MRR-Spark gap and its importance, determination of spark gap, factors on which it depends-EDM machine construction, table, work head, motors, spindle, dielectric tank. Flushing purpose and method of flushing-Safety precautions-Electro Chemical Machining (ECM) – principle – surface finish – application – advantages and limitations-Wire cut Electric Discharge Machining (WCEDM) – Introduction – principle – machining procedure – applications – advantages and limitations-Describe Electro Chemical wire cutting.

MODULE III

Abrasive jet machining – introductions, equipment, accuracy and surface finish, applications, advantages and limitations- Laser Beam Machining (LBM) – Introduction, equipment and machining procedure – accuracy – application advantages and limitations-ULTRASONIC machining (USM) –sketch, method, advantages and disadvantages-Water Jet Machining (WJM) – Introduction – principle of operation – equipment – advantages – limitations & applications.Plasma Arc Machining – Introduction – Principle – accuracy and surface finish –

advantages – limitations & applications. Electron Beam Machining – Introduction – Energy source – Principle – advantages – limitations – applications. Explosive fabrication – open type – closed type – multiple types.

MODULE IV

CNC/DNC with line diagram- description-Computer Integrated Manufacturing (CIM) -main activities under-CIM Computer Aided Design (CAD) –components of a CAD system. Computer Aided Manufacturing (CAM)- introduction – advantages of CAD and CAM application area of CAD and CAM-Flexible manufacture system (FMS) –flexibility and automation- basic computer control system – human labor-Types of flexible manufacturing system – flexible manufacturing unit (FMU) – Flexible Manufacturing cell (FMC) – flexible manufacturing transfer line (FMC) or (FTL) .

RP-Basic Principles of RP.RP Technologies (Liquid based, Solid based).Stereo lithography-Droplet Deposition manufacture-LOM-FDM-Practical applications of RP-Limitations of RP.

REFERENCES

1. Westerman tables
2. Workshop technology by Chapman CBS Publishers & Distributors
3. Technology of machine tools by Krar & Oswald
4. Production Technology by R K Jain Khanna Publishers
5. All about machine tools by Gerling New age International (P) Ltd.
6. Production technology by Hazra Choudhary – Indian Book distribution Company